

Summary

ML Research Engineer in robot learning and autonomous driving. Fine-tuned OpenPI VLA models, trained PPO manipulation policies in simulation, and shipped generative driving models at Cruise. Published at ICML, BMVC, and RecSys.

Work Experience

Scale AI

May 2025 - Present

ML RESEARCH ENGINEER

New York City, NY

- Fine-tuned π **vision-language-action models (openpi)** on open-source (Bridge) and Scale-collected manipulation datasets to measure data quality; **built visualizations to inspect flow-matching action denoising**.
- Built the **SFT cold-start** for multi-turn, tool-using **deep research agents** (distilled reasoning traces, fine-tuned Qwen3 8B/32B), then trained them with **GRPO reinforcement learning on SkyRL** and FSDP2, optimizing **verifiable rewards (RLVR)**.
- Co-authored **ICML 2026** paper with Scale's GenAI research team on trustworthy RAG retrieval evaluation via **semantic stratification**, yielding coverage guarantees and visibility into retrieval failure modes.

Robot Learning in Simulation

2026

PERSONAL PROJECT

- **Trained and evaluated PPO manipulation policies** in Robosuite/MuJoCo for cube lifting; **diagnosed reward-hacking failure modes** where policies optimized shaped rewards without completing the task.

Cruise

April 2023 - May 2025

ML/ROBOTICS ENGINEER

San Francisco, CA

- Trained and deployed **large generative driving models** in production: an auto-regressive trajectory model improved by 66% in inference latency via distillation, quantization, and a hybrid one-shot/auto-regressive scheme.
- Designed an **imitation learning** based structured ML model, scoring AV trajectories from multimodal perception + prediction inputs; built Transformers-based trajectory prediction on vectorized map/agent inputs.
- Implemented **low latency physics and math based cost functions in C++** to calculate distance/time based features during inference time, that would run on 100s of AVs in real time.
- Designed **custom loss functions** and distance-time based auxiliary losses for pretraining trajectory models, resulting in a 13% reduction in possible AV collisions and surfacing 90%+ of pedestrian edge cases.

Autonomous Vehicles Lab | Dynamic Scene Modelling

Oct 2021 - June 2022

PROF. HENRIK CHRISTENSEN, CONTEXTUAL ROBOTICS INSTITUTE UCSD

UCSD, La Jolla, CA

- Built 3D semantic maps for urban autonomous driving from Camera + LiDAR point cloud data, using ROS, rviz, and computer vision on the lab's self-driving vehicle.

Computer Vision Research Assistant

April 2020 - Nov 2020

BAIR LABS, UC BERKELEY

Remote

- Achieved **higher diversity in Generative models** by alleviating Mode Collapse in GANs caused by Catastrophic Forgetting, with an *increase in Inception Score by 0.3 and decrease in FID by 2.8*. **Accepted at BMVC 2021**.

Publications & Writing

- Coverage, Not Averages: Semantic Stratification for Trustworthy Retrieval Evaluation | *ICML 2026* | [link](#)
- Can Linguistically Related Languages Guide LLM Translation in Low-Resource Settings? | *LoResMT @ EACL 2026* | [link](#)
- Reward Hacking in RL Fine-Tuning of Foundation Models | *Blog, 2026* | [link](#)
- Overcoming Mode Collapse in Generative AI by Adaptive Multi-Adversarial Training | *BMVC 2021* (co-first author) | [link](#)
- A Synthetic Data Petri Dish for Studying Mode Collapse in GANs | *BayLearn 2020* (co-first author) | [link](#)
- Goal-driven Command Recommendations for Analysts | *Oral, ACM RecSys 2020* (first author) | [link](#)

Education

University of California San Diego

Sept 2021 - March 2023

MASTER'S IN COMPUTER SCIENCE | GPA : 3.9/4.0

San Diego, CA, USA

Indian Institute of Technology Kanpur

Aug. 2016 - May 2020

BACHELOR'S IN ELECTRICAL ENGINEERING, MINOR : MACHINE LEARNING | GPA : 9.0/10.0

Kanpur, India

Skills

Languages Python • C++ • CUDA • Shell

ML / DL PyTorch • JAX • TensorFlow • Transformers • CNNs • FSDP

RL & Policy Learning PPO • GRPO • imitation learning • VLA fine-tuning • flow matching • reward design

Robotics / Sim MuJoCo • Robosuite • ROS • OpenCV • LiDAR / point clouds